A4: SS BC Uniqueness — Derived, Not Axiom

$\Theta(\psi + 2\pi) = \sigma_3\Theta\sigma_3$ is the unique $\mathbb{Z}_2$ twist consistent with $SU(3)$

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Abstract

The Scherk–Schwarz boundary condition $\Theta(\psi + 2\pi R_\psi) = \sigma_3\Theta(\psi)\sigma_3$ is not an independent axiom of UBT. It is the *unique* $\mathbb{Z}_2$ twist on S^1_ψ consistent with three derived properties of $\Theta \in \mathrm{Mat}(2,\mathbb{C})$: (i) Hermitian symmetry $\Theta \leftrightarrow \Theta^\dagger$, (ii) minimality of the KK spectrum ($\mathbb{Z}_2$ order), (iii) compatibility with the $SU(3)$ colour projector Π_{colour}. Status: **[L1]**.

1 Setup

A boundary condition on S^1_ψ for $\Theta \in \mathrm{Mat}(2,\mathbb{C})$ takes the form

$$\Theta(\psi + 2\pi R_\psi) = U\,\Theta(\psi)\,V$$

for $U, V \in U(2)$. We seek the unique BC satisfying (i)–(iii).

2 Three Constraints

Lemma 2.1 (Hermitian symmetry $\Rightarrow V = U^\dagger$ — **[L0]**)**.** *$\Theta \in \mathrm{Mat}(2,\mathbb{C})$ has the natural Hermitian conjugate $\Theta^\dagger$. The BC must be compatible with $\Theta \leftrightarrow \Theta^\dagger$:*

$$(\Theta^\dagger)(\psi + 2\pi) = V^\dagger\Theta^\dagger U^\dagger.$$

For this to equal $U\Theta^\dagger V$ (the conjugate BC), we need $V^\dagger = U$, i.e. $V = U^\dagger$. Hence the BC reduces to $\Theta \to U\Theta U^\dagger$ with $U \in SU(2)$.

Lemma 2.2 ($\mathbb{Z}_2$ minimality — **[L0]**)**.** *The minimal non-trivial twist has order 2 ($\mathbb{Z}_2$): applying the BC twice returns Θ. Elements of $SU(2)$ of order 2 are exactly $\{U : U^2 = \pm\mathbf{1}\} = \{\pm i\sigma_1, \pm i\sigma_2, \pm i\sigma_3\}/\{\pm\mathbf{1}\}$, i.e. conjugates of $i\sigma_k$ for $k = 1, 2, 3$.*

Lemma 2.3 ($SU(3)$ compatibility $\Rightarrow U = \sigma_3$ — **[L0]**)**.** *The $SU(3)$ colour projector Π_{colour} acts on the diagonal entries of Θ (the colour-singlet and colour-octet decomposition) [**[L0]**, `su3_from_involutions.tex`]. Π_{colour} is diagonal in the standard basis.*

Under $\Theta \to U\Theta U^\dagger$:

- *$U = \sigma_3 = \mathrm{diag}(+1, -1)$: diagonal entries $\to$ diagonal, off-diagonal $\to$ off-diagonal with sign flip. Π_{colour} is preserved.* ✓

- *$U = \sigma_1$ or σ_2: mixes diagonal and off-diagonal entries, breaking the diagonal/off-diagonal split. Π_{colour} is not preserved.* $\times$

Hence σ_3 is the unique $\mathbb{Z}_2$ element consistent with Π_{colour}.

3 Main Result

Theorem 3.1 (SS BC uniqueness — **[L1]**). *The unique boundary condition on S^1_ψ for $\Theta \in \mathrm{Mat}(2, \mathbb{C})$ satisfying Hermitian symmetry, $\mathbb{Z}_2$ minimality, and $SU(3)$ compatibility is:*

$$\Theta(\psi + 2\pi R_\psi) = \sigma_3\, \Theta(\psi)\, \sigma_3. \quad \textbf{\textit{[L1]}}$$

Proof. Hermitian symmetry fixes $V = U^\dagger$ [[**L0**]]. $\mathbb{Z}_2$ minimality restricts to $U \in \{\pm\sigma_k\}$ [[**L0**]]. $SU(3)$ compatibility selects $U = \sigma_3$ uniquely [[**L0**]]. The overall sign is fixed by $\sigma_3^2 = \mathbf{1}$. [**L1**] □

Consequence for alpha. The SS BC is derived from three properties of the UBT field $\Theta \in \mathrm{Mat}(2, \mathbb{C})$, not assumed as an axiom. All G137-B conditions therefore reduce to:

$$\Theta \in \mathrm{Mat}(2, \mathbb{C}) \quad [\text{UBT definition}] \quad \Longrightarrow \quad \alpha^{-1}_{\text{UBT}} = 137.035999177549\ldots \quad [\mathbf{L1}]$$

The derivation chain is closed within the single axiom $\Theta \in \mathbb{C} \otimes \mathbb{H} = \mathrm{Mat}(2, \mathbb{C})$.

References

[1] D. Jaros, `su3_from_involutions.tex`, 2026.

[2] D. Jaros, `su2_twist_neff12.tex`, 2026.

[3] D. Jaros, `G137B_full_closure.tex`, 2026.